

Is AI the new internet?

Introduction

The introduction of the internet to the public in the early 1990s brought a transformation of communication and distribution of information globally. For organisations, this shifted operations from analog methods such as mail and fax, to email and instant messaging over a global network. This shift allowed companies to distribute and share larger amounts of data than ever before, at unprecedented speeds.

Just as the internet revolutionised communication and distribution, modern artificial intelligence is revolutionising cognition. Machine learning algorithms mimic the human process of learning by being 'trained' on massive datasets, which means that its users can augment themselves to also have access to its knowledge by proxy. Additionally, Agentic AI can be used to automate decision making, or act on behalf of users to achieve specific goals with minimal supervision.

With the advent of the world-wide-web, came the crash of the '*Dot-com bubble*' a couple years after, caused by an influx of internet-based startups and speculative investment into them. Some analysts have recently been speculating that AI may end up having a similar fate, since it has also spurred an exponential increase in startups and trillions of dollars of investment, with most of the top 10 biggest companies in the world being related to AI.

This essay explores how AI has created this trillion-dollar value, and how it can be sustained to avoid such a fate. While AI offers immense value by augmenting knowledge work, successful adoption requires proactive leadership and human-centric HR strategies to manage complex first and second-order effects, balancing productivity with the preservation of social cohesion.

AI's Unique Features and Strategic Value

Deterministic vs Probabilistic Algorithms

The way AI works is fundamentally unique compared to earlier workplace technologies such as ERP (Enterprise Resource Planning) systems. Earlier technologies rely on simpler rule based or deterministic algorithms, operating like rigid flowcharts where fixed inputs produce identical outputs. For example, an ERP system may have a pre-programmed rule of *'if stock drops below 10, order 50 units'*, which it will always follow.

Probabilistic models incorporate uncertainty, producing varied outputs based on probability distributions- this is what allows machine learning algorithms to learn, infer and make decisions, and deal with noisy, incomplete or ambiguous data. It works by mapping patterns in vast datasets into high-dimensional vectors, giving it the ability to identify complex patterns and relationships otherwise invisible to rule-based logic. This gives it the ability to infer and use context to calculate most probable outcomes of various situations, meaning it can generalise and handle scenarios it's not explicitly programmed for.

Agentic AI and Scalability

Because of its capabilities, modern AI is evolving from a passive tool that requires prompts for every action, into a system that can make autonomous decisions and pursue more complex goals independently or with minimal supervision. Unlike a standard LLM (Large Language Model) which will only answer a single prompt, Agentic AI can analyse the context and environment of a complex problem, figure out what steps to take, then independently execute a sequence of actions to achieve a desired outcome.

This means that, to some extent, an AI Agent can act as a counterpart for certain clerical or administrative tasks. For organisations, this means a new form of scalability, representing a shift in strategic value from 'augmentation' to 'autonomy'. A human manager can now manage a fleet of AI agents, each with distinct, simple workflows such as debugging code or managing customer inquiries; instead of a team of human employees who would be doing the same jobs.

The possibility of being replaced by AI is a threat to the existence of many junior or entry-level jobs (more on this in later sections of this essay). While agents can reduce cognitive load, in practice AI cannot and should not be entirely relied on due to its tendency to hallucinate and lack of authority or accountability within an organisation. A '*human-in-the-loop*' is still necessary to monitor the AI's logic, set goals, and intervene when it inevitably hallucinates or drifts from ethical guidelines. This means that each of a companies' employees are equipped with this powerful tool that augments them to automate simple, tedious clerical tasks, reducing their cognitive load and allowing them to work on more important tasks that the AI can't independently do reliably.

GenAI and 'The Jagged Frontier'

Traditional IT systems merely store, retrieve and process existing data, while Generative AI is capable of creating new content such as text, code or graphics (Hinds & von Krogh, 2024), which is one of its' most distinct features. This is especially impactful for departments that require graphical design or content creation such as marketing. It allows these teams to, for example, generate multiple iterations of a graphic then spend their time refining the best one, rather than spending hours rendering a single concept.

A workflow like this introduces the concept of the '*jagged frontier*' of AI. It excels at tasks such as generating lots of content to be used for inspiration, but fails at other more 'human' things like nuanced ethics or creative judgement- unlike previous deterministic technologies which

have consistent performance and results for the tasks they are programmed to do. This creates a ‘jagged’ line on the boundary between what AI can and cannot do (Dell'Acqua et al., 2025).

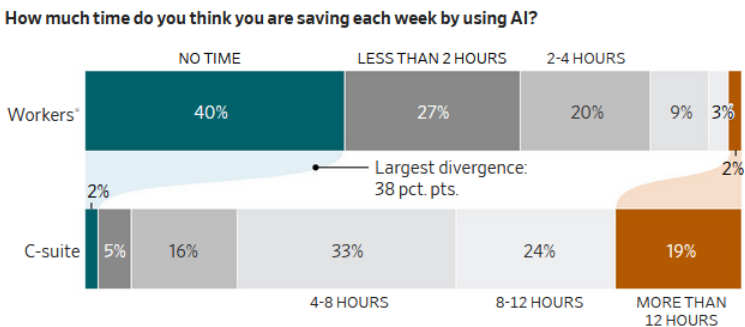
Consequently, the strategic value of AI-human collaboration is often not uniform, and ‘*skill biased*’. Research indicates that lower-level jobs often benefit more from AI augmentation than higher-level ones, acting as a ‘leveler’ that fills in gaps in technical knowledge for novices (Jia et al., 2024). For an expert on the other hand, it might act as a ‘sparring partner’ that challenges their assumptions, but its’ output still requires rigorous critical evaluation.

First and Second-Order Effects

AI undeniably has massive potential to accelerate progress and create value for everyone, no matter the industry or job position. However, the messy reality of implementation often comes with friction. This section will explore the theoretical and practical ripple effects of AI, and how it also has the potential to paradoxically hurt productivity, as well as ‘hollowing out’ the labour market in the long term.

First-Order Effects: The Paradox between Efficiency and ‘Mental Atrophy’

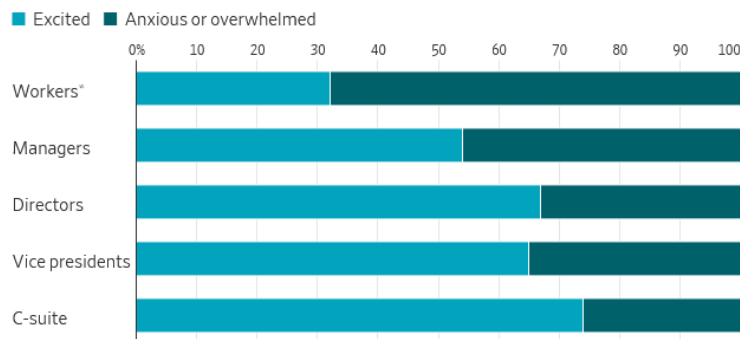
Initially, AI can successfully automate tedious, low stakes tasks such as data entry, writing everyday emails, writing and debugging code, etc. However in reality, there seems to be a disconnect between executives’ and employees’ opinions on AI’s efficiency. A recent Wall Street



Journal (2026) study found that while executives and CEOs expect massive time savings, employees report that ‘cleaning up’ AI’s errors often takes just as long as doing a task manually.

There also seems to be a correlation in workers' optimism in AI, and their hierarchical power within the company (Ellis, 2026). Workers tend to use AI most commonly as a research assistant

Which of the following best describes how you feel about AI?



or Google replacement or for generating drafts, rather than more complex tasks like data analysis or code generation, like some executives may wish their employees to be doing.

While AI helps learning, the daily grind of correcting it causes fatigue (Shao et al. 2024). Workers that don't bother 'cleaning up' AI's errors may fall into 'automation bias'- because its output looks plausible and hallucinations often sound confident, employees may uncritically accept AI outputs to save effort. Automating an email reply saves time, but removes the 'human touch' required for sensitive client relationships, for example. A generic AI-written response to an angry client can cause immediate reputational damage to the company- then an additional dilemma of accountability occurs, especially if the response was sent automatically by an AI agent, without a human-in-the-loop to click the 'send' button. This is an example of the 'Borg Effect' described by Fuegener et al. (2021), where humans rely on the AI and their unique individual knowledge 'atrophy' and their decisions converge, reducing the 'wisdom of the crowd'.

Another unintended consequence of this is 'homogenisation', where things from different companies start to look and sound the same because they used the same foundation model. For example, you can often tell if the website of a new start-up was 'vibe-coded', as they often share the same visual habits, layouts, styles and colours. AI-generated marketing material is also often immediately pointed out by the public, as it usually has a certain recognisable 'style' to it.

With anything new and controversial, a common first-order effect is backlash and fear- a significant proportion of the global public are 'anti-AI', and according to the WSJ study, 40% of the 5,000 white-collar workers they surveyed said they would be fine never using AI again. It is possible this friction is due to the fact that AI is still relatively new, and employees are still getting used to using it or haven't had proper training from their bosses, which is something that could be naturally smoothed out in the long term with proper management.

Second-Order Effects: The AI-Powered Gig Economy

Mercor is an AI-powered labour marketplace, originally providing paid opportunities for various simple AI training, data annotation and ontology jobs. Recently, these opportunities have expanded- a particular listing that struck me personally was 'radiologist'. If even high skill talent like this can be 'unbundled' into specific tasks, this suggests massive future implications for hiring and firing habits across the entire labour market. By doing this, Mercor's CEO is using AI to redefine who gets to work, how teams are built, and how talent is used and distributed in a global, automated economy (TechCrunch, 2025). Traditional, full time jobs may become globally 'gigified' which shifts the supply curve and increases scheduling flexibility which is of great strategic value for the hiring company, but reduces job security, benefits and control for the worker (Mohlmann et al., 2021).

Redistribution of Expertise and Power, and Disruption of the 'Social Fabric'

If this sort of Mercor model dominates, the workplace may become a transactional network of isolated tasks, rather than a community. The efficiency of a high-turnover AI-powered HR organisation threatens the formal structures and informal 'social fabric' between employees where long-term relationships are usually built, moving from 'coexistence' to simply temporary sprints of 'co-creation'.

This may also redistribute expertise and power upwards. If AI automates the ‘drudgery’ that junior staff usually do, they lose the training ground where they used to learn such tasks, and they never get the chance to develop the tacit knowledge required to become seniors. The organisation gains immediate efficiency, but hollows out its long-term talent pipeline, and becomes top-heavy with experts (Lebovitz et al., 2021).

Managing the Adoption of AI Augmentation

In a world where the technological environment is evolving at an accelerating rate, agility is critical for survival. While the technological potential is valuable, the organisational first and second-order risks are equally high, and socio-technical change must be managed responsibly so that human-AI collaboration is sustainable. Generally, this means a strategic shift from a technology-first approach of simply introducing AI wherever possible, to a human-first approach of redesigning employees’ workflows and augmenting them- and crucially, keeping them trained so they know how to use it effectively.

Agile Leadership and Accountability

The unpredictable nature of AI tools means static, long-term roadmaps become obsolete very quickly. Instead an iterative approach should be used, where teams experiment with different tools and ideas in controlled environments, learn from each iteration, and adapt tactics and strategies on the go (Retkowsky et al., 2024). This agility allows specific risks to be spotted before they scale into systematic failures. This agility must also be paired with accountability to reduce the risk of automation bias, the ‘Borg effect’, and the dilemma of responsibility especially associated with agentic AI. The stakes and consequences of all tasks must be assessed, so that the locus of accountability for the outcomes of sensitive tasks stays within the grasp of a human.

Preserving the Talent Pipeline

Certain clerical tasks that junior staff usually do can already be completely automated. To prevent the junior end of the talent pipeline from becoming obsolete, one strategy could be to embrace 'intentional inefficiency', where efficiency is not a primary metric. This could be in the form of providing training grounds where juniors can still learn the tacit and explicit knowledge and skills required to upskill themselves in a sandbox. Entry-level job descriptions could also be redesigned to keep a human in the loop so that these jobs are augmented not replaced by AI, prioritising long-term social responsibility over maximised efficiency and investing into a sustainable future (Raisch & Krakowski, 2021). The 'jagged edge' of AI should be evaluated to understand its' strengths and weaknesses, and a human must always be there to oversee and fill in for the weaknesses.

AI Literacy

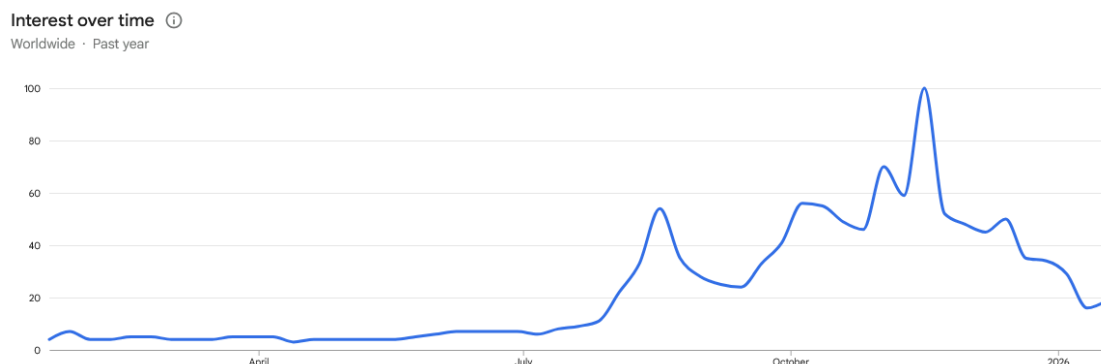
Misunderstanding often breeds fear and distrust. If executives want their employees using AI effectively, proper AI literacy training must be implemented alongside it. Employees should understand that it is a probabilistic tool that requires validation, rather than a deterministic truth machine. Proper prompt engineering is also an important yet often overlooked skill that can be used to fine-tune a desired output, reducing 'clean-up' of generated errors. AI adoption should also be framed as a means of augmenting capacity rather than replacing headcount to establish psychological safety regarding job security- potentially by encouraging the use of LLMs as a 'sparring partner', or rewarding identification of AI errors. This reduces fear and distrust, and cultivates a culture that is AI-literate, not just AI-equipped (Stark & Broeck, 2024).

Ethical Guardrails, and Preserving the 'Social Fabric'

Appropriate ethical guardrails that prioritise social cohesion must be put in place to prevent the hyper-efficiency of labour marketplace models such as Mercor from eroding the social fabric. Leaders should invest in something AI cannot do: culture and connection. Bankins et al. (2024) suggests workflows should be intentionally designed to require human-to-human collaboration as much as possible, especially for higher-stakes decisions. In practice, this means that time saved by AI should be used for building relationships with mentorship and creative brainstorming, rather than simply for doing more work, ensuring organisations remain a place where humans collaborate and create value for society, rather than a platform for automated tasks.

Conclusion

To prevent a bubble from inflating with speculation and bursting, real, tangible value must be created and sustained. Modern AI offers immense value by separating intelligence from human biology and into a machine. The AI revolution has many similarities to the internet revolution, and people back then had many of the same concerns of being replaced by computers. Decades later, despite the Dot-com bubble crashing on the stock market, the internet's social and economic value still thrives.. By using similar frameworks and guardrails with AI that we used then- giving technology into the hands of the people and teaching them how to use it as a catalyst- organisations can ensure sustainable, long-term resilience and value. Fears of an AI bubble have also slowed down in recent months, suggesting that appropriate strategies may already be being put in place. Maybe the idea of an 'AI bubble' is a bubble in itself (Google Trends, 2026).



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Appendix

Word count without references or appendix: 2498.

Total word count: 3447.

AI use declaration

This content is original, all material has been written by myself and no parts of the assignment were AI generated, and all sources are correctly and properly acknowledged.

I used Gemini 3 Pro to:

- help me sift through sources and find relevant ones to use
- Bounce my own ideas off of
- Give me ideas for overall structure of essay
- Using critical thinking, sanity checking, fact checking, etc

Essentially, rather than AI 'replacing' me and doing the work for me, i used AI to augment myself and collaborated with it in a way, in similar ways that one can collaborate with fellow students, or someone knowledgeable in the topic- or employing someone to sift through sources for me to save me time.

This aligns with the AI use guidelines in the brief:

'For this assignment you can use AI to refine ideas and research materials but 100% of the text in the submitted assignment must be entirely written and developed by the student. Or in other words, no text in the assignment can be written by any AI tool.'

Prompts used:

- ive attached an assignment brief for a 2500 word essay. it says in the brief: ‘...’ which means there are essentially 3 questions to answer, so about 700-800 words per question. give me an essay plan for this uni assignment and some relevant talking points, along with relevant sources to refer to (since it says 10-15 are needed). I have also been given from my lecturer a folder of around 70 pdfs as potential readings of papers on AI- can you process this amount of pdfs if i share them all with you?
- here's a screenshot of all the names of the sources (i guess with some you cant see the full file name, you can still try figure out if they're relevant)
- Here are some notes and ideas I have currently written down. Start planning an essay structure- headings and subheadings about topics to talk about, roughly following the points of the unfinished notes i have provided.

Removed Content

This paragraph was originally an introduction to the ‘AI’s Unique Features and Strategic Value’ section, but removed due to potentially being irrelevant to save word count:

‘Artificial intelligence is not new. Its foundational ideas and early research began in the 1940s with figures like Alan Turing who formulised computer science theory, and explored the idea of ‘thinking machines’. These earliest computers were simply mathematical algorithms for specific problems. Computers later developed into digital machines powerful enough to be programmed and run more complex algorithms capable of computing bigger problems, such as running different kinds of programs simultaneously. Over time, both hardware and software became more powerful and advanced, eventually evolving into AI models being hosted in datacenters across the world.’